

TRANSMISSION CORPORATION OF ANDHRA PRADESH LIMITED

QUALITY ASSURANCE PLAN

&

CHECK FORMATS

FOR 220KV & 132KV TRANSMISSION LINES AND

SUBSTATIONS CONSTRUCTION

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PREAMBLE

During the Quality Control Review Meeting on 03.08.2007 the Chairperson and Managing Director/ APTransco has instructed to prepare the quality assurance plan for erection of 132 KV and 220 KV lines & substations for implementing in APTransco.

Accordingly, the Quality Assurance Plan is prepared for erection of 132 KV and 220 KV line & substations and approval accorded by technical committee for implementing in APTransco.

The following issues were deliberated to ensure quality during execution of works:

1. Check Survey

i) Verification of line deviation angle and bisection for proper location of towers.
ii) Verification of meeting statutory clearances.
iii) Check spans on either side of the tower location.
iv) Mark distances and angles from angle point at 3 conspicuous places to identify the angle point location in case of displacement of angle point peg.
v) At peg marks GPS marking should be done. This ensures correct position of peg even if it is removed, as the longitudes and latitudes are recorded.

2. Soil Investigation

i) Identify 75 locations for performing SPT.
ii) SPT to be performed at all angle points, at locations where soil strata changes, railway and national/state road crossings, power line crossings and special locations limiting to the 75 locations as provided in the contract.
iii) Instead of identify 75 locations, it is better to perform soil investigation for every KM of soil. Investigation is done by way of ERM method, time is saved and soil investigation can be done without disturbing standing crops.

3. Pit Marking

i) Mark position of all the four pits on the ground as per pit marking drawing for standard type of foundation and note levels of the center of each leg pit w.r.t center peg of the tower.
ii) Assess volume of bench cutting or use of hill side extension revetment if required and take approval of the Superintending Engineer /Chief Engineer.

4. Foundation Classification

i) One of the foundation pit to be used as a trial pit and excavated as per normal soil dry type foundation.
ii) Observe following for classification
- Type of cultivation surrounding tower and whether encasement of stub required due to surface water.
- Type of strata and depth of each layer.
- Water table/level.
- Maximum water table in the near by village well.

iii) Check and record soil classification and foundation classification as per relevant clauses of the agreement and excavate pit as per approved drawing for the type of foundation classification. (Extract of Technical Specification is enclosed).

4.1 CLASSIFICATION OF FOUNDATIONS:

Classification of soil shall be made according to IS : 200 (Part I) 1974 for footing cast in open pits. The foundation designs for various types of Foundations as given below shall be adopted in the construction of all 132 kV & 220 kV Transmission Lines and shall be based on the following instruction. The Foundation Drawings as finalized by the Civil Engineering Wing for various soil conditions suitable for different Types of Towers being deployed in the construction shall be followed by all field officers strictly.

Sl.No.	**Soil at the Location of Tower**	Classification of foundation to be adopted.
1.	Where normal dry cohesive or non-cohesive soils are met.	Dry
2(a).	Where sub-soil water is met at 1.50 meters or more below the ground level in normal soil	Wet
(b)	Where surface water could stand for long period with water penetration not exceeding 1.0 m below ground level (e.g. Paddy fields, Sugarcane fields)	
3(a).	Where sub-soil water is encountered between 0.75 meter and 1.50 meters depth below ground level.	Partially Submerged / Partially Black Cotton
(b)	Where top layer of Black Cotton soil extends up to 50% of the depth with normal soil thereafter	
Note – 1	Where soil is clayey type, not necessarily black in colour, which shrinks when dry, swells when wet.	
4(a)	Where sub-soil water Table is within 0.75 m depth from ground level.	Fully Submerged – Black Cotton
(b)	Where top layer of Black Cotton soil exceeds 50% and extends up to full depth or is followed by normal soil.	
(c)	Where top layer is normal soil up to 50% of the depth but the lower layer is a Black Cotton soil	
Note – 2	In case of items 4(a), (b) & (c) above, the concerned Superintending Engineer (Civil) shall personally inspect and certify on soil classification and shall not delegate this responsibility to any sub-ordinate officers.	
5(a)	Where decomposed or fissured rock, hard gravel, kanker, limestone, laterite or any other soil of similar nature is met.	Fissured / soft rock
(b)	Where top layer of normal dry soil extends up to 85% of the depth followed by fissured rock without presence of water.	
6(a)	Where Hard rock is encountered at 1.5 m or less below ground level.	Hard Rock
(b)	Where Hard rock is encountered form 1.5 m to 2.5 m below Ground Level though Top layer is of normal soil	
(c)	Where chiseling, drilling and blasting is required for excavation.	
Note – 3	While classifying foundation of Wet, Partially Submerged / Partially Black Cotton, Fully Submerged / Black Cotton foundations mentioned above, the worst conditions may be considered and not necessarily the conditions prevailing at the time of inspection. For instance, there are areas where sub-soil water rises when canal water is let-out in the field raising sub-soil water to a considerable degree. Similarly the effect of monsoon or when the nearby reservoirs are full should be considered and not the conditions prevailing in open season or summer when work is carried out normally. In all such cases, the Chief Engineer (Civil) shall personally inspect and certify the requirement of higher classification on case to case basis.	

Note – 4	For critical locations of cut-point / Angle Towers, the classification of Tower Foundations may be at next higher type for ensuring safety of the transmission line as likely to be subjected to variable forces from time to time on account of change in season and climatic conditions. This shall be decided by the concerned Superintending Engineer (Civil) on case to case basis on personal inspection.
Note – 5	In case of highly collapsible & loose soils and river crossings, special design for foundations are to be made by Civil Engineering Wing for adoption on case to case basis.
Note –6	At locations subjected to water logging, particularly in paddy fields, ponds, marshy areas, etc., the chimney of all stubs to be raised to 600 mm over the normal chimney being adopted in other cases. This shall be done after complete erection of towers as a separate item for identified locations.

In addition to the above, depending on the site conditions more varieties of foundations may be introduced suitable for intermediate conditions under the above classifications to effect more economy.

5. Form Work

Verify form work dimensions, rigidity, adequacy of tightness to prevent loss of cement water from concrete, ensure form surfaces in contact with concrete wetted and sprayed with fine sand/treated with black oil or waste oil/mould releasing agent every time before use.

6. Stub setting and Stub setting template

i) Recheck the depth of the pits w.r.t center peg level and pit dimensions as per approved foundation classification.
ii) Check under cutting as per approved drawing.
iii) Position fully assembled and bolt tightened stub setting template with supporting jacks placed on firm ground and level the template.
iv) Check placing a green concrete block of thickness equal to the clearance between the stub end and concrete to mitigate settling of template.
v) Check height of template above the center peg level.

7. Reinforcement

i) Check reinforcement dimensions as per bar bending schedule. Tie reinforcement at alternative crossing points with suitable binding wire.
ii) Verify using chair and gitties to ensure proper spacing in two layers reinforcement and minimum cover of 50 mm on the bars respectively.
iii) Before commencement of concreting ensure sufficient quantity of chairs and gitties are fabricated and made available at stores for use at locations.

8. Concrete Mixing

i) Ensure using coarse aggregate (metal) and fine aggregate (sand) as per IS-383.
ii) Check coarse aggregate for flakiness, soft stone and laminar shape.
iii) Check cement for approved brand.
iv) Ensure proper mixing of concrete as per the guide lines.
v) After mixing and before placing the concrete in the foundation pit, take samples for checking compressive strength and slump test of concrete.
vi) Engage ready mix where even possible to enable execute fast work.

9. Placing of Concrete

i) Check providing lean concrete (M 10) base 100 mm thick at bottom of the excavated pit before placing reinforcement.
ii) Use pointing/poking rod and vibrator for compaction of concrete ensuring vibrator head does not come in contact with steel, hardened concrete etc.,

10. Stripping of Form Work

Ensure stripping formwork after about 24 hrs and check for any damage, honey combing and do necessary rectification.

11. Back Filling

i) Ensure back filling on foundation in layers and ensure proper consolidation.
ii) Ensure all top soils to be placed at the surface in case of towers located on cultivated land.
iii) Excess earth available over and above ground level (after consolidation) should be transported away from the site.

12. Curing

Ensure proper curing by keeping adequate quantity of water and storing arrangement available at site.

13. Earthing of Tower

i) Ensure tower footing resistance not to exceed 10 Ohms.
ii) Use pipe earthing or counter poise earthing as per approved drawings to ensure minimum tower footing resistance.

14. Formats & QAP is revised with reference to the discussions and the revised formats & QAP are enclosed herewith. The Superintending Engineers should get the formats printed and bounded as registers and to ensure, they are distributed to Section Officer level.

15. All the foundation drawings, and other instructions shall be distributed upto Section Officer level.

16. PENALTIES IMPOSED FOR DEFICIENCEIES IN QUALITY OF WORKS:

During execution of works , if any deficiencies in quality of works is found in deviation to the specification / Agreement, a minimum penalty of Rs 5000 to Rs 30,000/- shall be levied for deficiencies as per each category as mentioned below.

Category I : Not using (i) prescribed shoring, shuttering and dewatering equipment, (ii) measurement boxes, (iii) Form boxes for different types of foundations and steel measuring boxes, (iv) not providing adequate number of chairs to the steel reinforcements, (v) not carrying out back filling and compaction of the foundation pits in layers and leveling the tower footings properly, (vi) not ensuring that the excavated earth is dumped at least 2 meters away from the pit etc., and (vii) not providing copings to the tower legs / stubs,(viii) not providing water tanker, Earth rammers/Earth vibrators.

Category II : Use of reinforcement steel without ISI marking, not using vibrators for effective consolidation of the concrete during foundation works, not using proper templates for firmly keeping the stubs in position when templates are supplied by the contractor & improper fixing of stubs, non ensuring of tower verticality, use of rusted stubs and tower parts if supply is by the contractor, non painting of butt joints and rusted stubs with zinc rich paint immediately after erection of the towers & stringing, non fixing of earth flat to the stub, non deployment of technical personnel for supervision of works by the contractor. Also not fixing of vibration dampers firmly, repair sleeves wherever necessary, not properly fixing of arcing horns both tower side and line side and bird guards etc., as per specification.

Category III : Use of improper grade / quality of raw material like H.B.G. metal, water and sand for concreting, using clogged and / or lump / clotted cement for concreting, not ensuring proper curing for foundation concrete, not ensuring that all the members of the tower are placed in position and firmly fixed with bolts and nuts immediately after erection of tower, not ensuring that Half round seam welding of the nuts before stringing of the line.

For the deficiencies in the quality of works noticed by the Engineer , penalties shall be levied as given below:

Type of category	First instance(Rs)	Second instance (Rs)	Third instance (Rs)
Category I	5,000 /-	10,000/-	15,000 /-
Category II	7,500 /-	15,000/-	22,500 /-
Category III	10,000 /-	20,000/-	30,000 /-

In the event of fourth instance of noticing the deficiency of quality of works in any of the above categories, the bidder shall be debarred from participating **in** future tenders for a period of one year.

The bidder shall give a specific undertaking accepting this clause.

TRANSMISSION CORPORATION OF ANDHRA PRADESH LIMITED
QUALITY CONTROL & INSPECTION DEPARTMENT

STANDARD FIELD QUALITY PLAN FOR 220/132 KV LINES & SUB-STATIONS

Section: SURVEY & SOIL INVESTIGATION

Sl. No.	Component Operation & Description of Test	Sampling Plan with Basis	Reference Documents and Acceptance Norms	Testing Agency	Remarks	Checking Authority
1	**DETAILED SURVEY & ALIGNMENT**					
(a)	Field Survey/Walkover Survey	100%	Technical specification Geogrphical Map, Route alignment, Route Plan & Measurement Schedules	Contractor	Counter check and approval by APTRANSCO	AEE/EE/ SE
(b)	Plotting of Route	100%	Technical specification Field book, Geogrphical Map, Route alignment, Route Plan	Contractor	Counter check and approval by APTRANSCO	AEE/EE/ SE
©	Profiling	100%	Technical specification Field book, Graph sheets, approved route alignment.	Contractor	Counter check and approval by APTRANSCO	AEE/EE/ SE
(d)	Tower Spotting	100%	Technical specification Approved Tower spotting data & Sag Template, Field Book, approved route alignment	Contractor	Counter check and approval by APTRANSCO	AEE/EE/ SE
(e)	Tower Schedule (Proforma to be approved by (APTRANSCO)	100%	Technical specification Approved Route alignment & route profile tower schedule, sag template, tower spotting data	Contractor	Counter check and approval by APTRANSCO	SE/CE
2	**Check Survey**					
(a)	Bisection of Angle/Accuracy of Alignment	100%	Technical specification IS:5613, and other relevant standards. Approved Route Map & Route Profiles, Tower/line Schedule and Sag Template etc.	Contractor	Counter check and approval by APTRANSCO for approval	AEE/EE/ SE
(b)	Check for profile levels and electrical clearances	100%	DO	Contractor	Counter check and approval by APTRANSCO for approval	AEE/EE/ SE
©	Check for span marking and lengths	100%	Technical specs IS:5613 and other relevant standards Apprd. Profile drgs. Tower schedule, tower spotting data.	Contractor	Counter check and approval by APTRANSCO	EE/SE
(d)	Check for tower type& position as per site condition	100%	DO	Contractor	Counter check and approval by APTRANSCO	EE/SE
(e)	Estimation of benching &Revtment volumes (as per site conditions)	100%	Technical specification IS:5613 and other relevant standards Apprd. Tower leg location counters, approved profile	Contractor	Counter check and approval by CE/400KV/L&SS/ APTRANSCO	Benching by EE/Field Revetment by SE
(f)	Final Profile & tower schedule	100%	Modified profile drawing and amended tower schedule	Contractor	Counter check and approval by APTRANSCO	EE/SE/CE

Sl. No.	Component Operation & Description of Test	Sampling Plan with Basis	Reference Documents and Acceptance Norms	Testing Agency	Remarks	Checking Authority
3	**Soil Investigation**					
A	**At Normal Locations**					
(i)	Borelog/Trial Pit for obtaining details/types of soil encountered. Including gradation of samples at specified levels, ground water levels etc. (at center of tower)	At locations as indicated by APTRANSCO	Technical specification IS:1498, 1892, 2720, 4464, 6935, SP-36 & and other relevant standards	Contractor/ APTRANSCO approved Agency	Counter check and approval by APTRANSCO The locations are to be identified and approved by APTRANSCO before carrying out soil investigation	AEE/EE/ SE
B	**Soil Resistivity**	One Location in a stretch of 2 to 3 Km or at locations approved by APTransco	IS:2131, IS 2720 and Technical Specifications	Contractor APTRANSCO approved agency	Counter check and approval by APTRANSCO	EE/SE
C	**Test on Soil and Rock Samples**				Approval by APTRANSCO	
C1	**Test on soil samples**					
i	Visual and Engineering classifications	DO	Technical specification IS:5613, IS:1498 and other relevant specifications			AEE/EE/ SE
C2	**Tests on Rock**	At locations as indicated by APTRANSCO	Technical specification IS:2131, IS:2720, IS:4078, 4474 and other relevant standards	Contractor APTRANSCO approved lab	Approval by APTRANSCO	AEE/EE/ SE
i	Visual Classifications		Technical specification IS:1498, 5613 and other relevant standards			
4	**Selection of Site for Material Storage**	100%	Technical specification, utility/industry storage document Wooden sleepers, wooden planks, tarpaulin, steel sheets, bricks, Barbed wire for fencing etc site shall be free from termites	Contractor	Approval by APTRANSCO	EE/SE
5	**Checking of Foundation Materials**					
A	**Cement**				Supply source to be approved by AP Transco	
(i)	Fineness	For each source MTC shall be furnished for every lot of 200 MTs and part thereof	APTRANSCO Technical specification IS:269, 456, 1489, 4031, 8112, 12269 and other relevant standards	Manufacturer works/ APTransco approved test lab/ contractor	Review and approval of manufacturers test certificates (MTCS)/laboratory test results by APTRANSCO	AEE/EE/ SE
(ii)	Compressive Strength	DO	DO	DO	DO	DO
(iii)	Initial & Final setting tim	DO	DO	DO	DO	DO
iv)	Soundness	DD	DO	DO	DO	DO
(v)	Chemical composition of Cement	DO	APTRANSCO Technical specification IS:269, 456, 1489, 4031, 8112, 12269 and other relevant standards	Manufacturer	Review of manufacturers test certificates (MTCS)/laboratory test results by APTRANSCO	SE/CE

Sl. No.	Component Operation & Description of Test	Sampling Plan with Basis	Reference Documents and Acceptance Norms	Testing Agency	Remarks	Checking Authority
B	**Coarse Aggregates**	One sample per lot of 200Cum or part per each source for each size`	Technical specification IS :383, IS:2386 and IS:2430 & other relevant standards	APTRANSCO Approved Lab	Each source will be approved by APTRANSCO review and acceptance of test result by AP Transco	AEE/EE/ SE
(i)	Determination of Particle size (sieve analysis)	One sample per lot of 200Cum or part per each source	Technical specification IS :383, IS:2386 and other relevant standards	Contractor/ APTransco approved lab	Each source to be approved by APTransco	AEE/EE/ SE
(ii)	Flackiness Index	DO	Technical specification IS :383, IS:2386 and other relevant standards	Contractor/ APTransco approved lab	DO	AEE/EE/ SE
(iii)	Crushing Valve	DO	Technical specification IS :383, IS:2386 and other relevant standards	Contractor/ APTransco approved lab	DO	AEE/EE/ SE
(iv)	Specific gravity	DO	DO	DO	DO	DO
v)	Bulk Density	DO	DO	DO	DO	DO
vi)	Absorption Value	DO	DO	DO	DO	DO
vii)	Moisture Content	DO	DO	DO	DO	DO
viii)	Presence of deleterious materials	DO	DO	DO	DO	DO
C)	**Fine Aggregates**	One sample per lot of 200Cum or part per each source	Technical specification IS :236, 383, 456, 2386, 2430 and other relevant standards	Contractor/ APTransco approved lab	Each source will be approved by APTransco	AEE/EE/ SE
(i)	Gradation/ Determination of Particle Size (Sieve analysis)	DO	Technical specification IS :236, 383, 456, 2386, 2430, 4031 and other relevant standards	Contractor/ APTransco approved lab	Each source tol be by review and acceptance of test result by APTransco	AEE/EE/ SE
(ii)	Specific gravity and density	DO	Technical specification IS :236, 383, 456, 2386, 2430, 4031 and other relevant standards	Contractor/ APTransco approved lab	DO	AEE/EE/ SE
iii)	Moisture Content	DO	DO	DO	DO	DO
iv)	Absorption Value	DO	DO	DO	DO	DO
v)	Bulk Density	DO	DO	DO	DO	DO
vi)	Presence of deleterious materials	DO	DO	DO	DO	DO
D)	**Water**					
(i)	Suitability (Potable water is considered suitable for concreting)	One sample for source	Technical specification IS 456, IS:3025 and other relevant standards	Contractor	Each source to be approved by APTransco	AEE/EE
(ii)	Cleanliness (Visual Check)	100%	DO	DO	DO	AEE/EE
iii)	P.H. Value	One sample per source	DO Min-6 and Max-8	Contractor/ APTransco approved test lab	DO	AEE/EE
E)	**Reinforcement Steel**					
(i)	Identification & Size	Random	Technical specification and approved drawings & IS 432, 1139 & 1786 and other relevant standards	Contractor/ APTransco approved test lab	Review and approval of manufacturers test certificates (MTCS)/laboratory test results by APTRANSCO	AEE/EE/ SE

Sl. No.	Component Operation & Description of Test	Sampling Plan with Basis	Reference Documents and Acceptance Norms	Testing Agency	Remarks	Checking Authority
(ii)	Strength (Tensile test, Yield stress/proof stress)	For steel as per IS 1786 under 10mm one sample per each 25 MT or part thereof 10mm-16mm one sample per each 35 MT or part thereof. Over 16 mm 1 sample per each 45 MT or part thereof.	DO	Contractor/ APTransco approved lab	DO	AEE/EE/ SE
iii)	Percentage elongation		DO	DO	DO	AEE/EE/ SE
iv)	Bend Rebend Test		Technical specification IS: 432, 1139, 1786 2502 and other relevant standards	Contractor/ APTransco approved lab	DO	AEE/EE/ SE
v)	Reverse bend test for HDS Wires		IS:432, and other approved specification	Contractor/ APTransco approved test lab	DO	EE/SE
vi)	Chemical analysis test	One sample per source	IS:432, IS:1139, IS:1786 and specification	Contractor/ APTransco approved test lab	DO	AE/EE/SE
F	**Earthing material**					
(i)	Visual identification	100%	Technical specification IS: 5613 and other relevant standards	Contractor	Approval by APTRANSCO	CE/Const.
(ii)	Bill of material & installation	100%	Technical specification approved drawings & IS 5613	Contractor	DO	EE/SE
6	**Tower Excavation**					
A)	**Before Excavation**					
(a)	Checking of pegs condition as per line alignment	100% Every Location	Approved line alignment/profiles Tech. Specification. IS 4091, 5613, CBI & P Publication No.268 and other relevant standards	Contractor	Counter check and approval by APTransco	AE/AEE
(b)	Checking of pit marking as per drawing and RI	100% Every Location	Technical specification, IS:4091, 5613 & other relevant standards Approved, profile excavation and foundation drawings	Contractor	Counter check and approval by APTransco	AE/AEE
©	Trial Pit	DO	DO	DO	DO	AE/AEE
(d)	Classification of soil	DO	DO	Joint inspection by APTRANSCO & contractor	Approval by APTRANSCO	AEE/ EE/ SE
(e)	Pit marking	DO		Contractor	Approval by APTRANSCO	AE/AEE
(f)	Pit levels with reference to center peg levels	DO	DO	DO	DO	AE/AEE
(g)	Depth of pits as per center peg level (estimate)	DO	DO	DO	DO	AE/AEE
(h)	Benching (Where required)	Where required	Technical specification IS 5613, approved Foundation leg contour Profiles and other relevant standards	Benching implements/ Contractor	Approval by APTRANSCO	AE/AEE
B	**Excavation**					
(a)	Dimensional conformity	Each Location	Technical specifications IS:3764, 408, 4091 &5613, SP:16 and other relevant standards Approved excavation Drgs./ Specification	Pit excavation tools/ contractor	Approval by APTRANSCO	AE/AEE

Sl. No.	Component Operation & Description of Test	Sampling Plan with Basis	Reference Documents and Acceptance Norms	Testing Agency	Remarks	Checking Authority
(b)	Verticality & squareness of each pit	Each Location	Technical specification IS: 3764	Joint inspection by APTRANSCO	Approval by APTRANSCO	AE/AEE
©	Verification of classification of soil type and type of foundation	DO	DO	DO	DO	AE/AEE/ EE/SE
(d)	Shoring and Strutting (Check durability strength & soundness of staging, fonts adequacy and specific livels)	As required	Technical specification applicable state/ local body safety regulations	Steel sheets, wooden planks stuts etc/ contractor	Coutner check by AP Transco	AE/AEE/ EE
(e)	Dewatering (Where required)	As required	Tech. Specification. IS:456 and other relevant standards	Diesel engine pumps. Etc. contractor	Coutner check by AP Transco	AE/AEE
C)	**Stub & Template**					
(a)	Identification & Assembly	100% Every Location	Technical specification IS: 5613. Approved drawings of stub template and foundation drawing	Stub template, water level tapes, sand bags, Jacks etc/ Contractor	Counter check by APTransco	AE/AEE
(b)	Stub Setting	100% every location	DO	Green concrete pads, adjustable Jacks, Sand bags etc. contractor	DO	AE/AEE
©	Template levels width & diagonal	100% every location	DO	DO	DO	AE/AEE
(d)	Ground Clearance	DO	DO	DO	DO	AE/AEE
(e)	Stub clearance	DO	DO	DO	DO	AE/AEE
(f)	Tightening of bolts and muts for template, stubs & cleats	DO	DO	DO	DO	AE/AEE
D)	**SHUTTERING (Form work)** Fixing of form boxes check for identification dimensions parallisation and squareness and equidistance from stubs	100% All collapsible location	Technical specification IS 456 and other relevant standards approved drawings	Joint inspection APTRANSCO & contractctor	Counter check and approval by APTRANSCO	AE/AEE
E)	**Concreeting**	100%	Technical specification IS 456 and other relevant standards & approved drawings	Joint inspection APTRANSCO & contractctor	Counter check and approval by APTRANSCO	AE/AEE
(a)	Control, Concrete mixed design		Technical specification IS 456 and other relevant standards & approved drawings	Joint inspection APTRANSCO & contractctor	Counter check and approval by APTRANSCO	AE/AEE/ EE
(i)	Identification of material and water supply sources	100%	Technical specification IS 383,456 IS:3025, 3550 and other relevant standards	Contractor	Counter check and approval by APTRANSCO	AE/AEE/ EE
(ii)	Determination of characteristics of aggregates, cement, water etc.	100%	Technical specification IS 269,383,456, 3125, 3550 and other relevant standards	Contractor/ APTRANSCO approved test laboratory	Counter check of test results and approval by APTRANSCO	AE/AEE/ EE

Sl. No.	Component Operation & Description of Test	Sampling Plan with Basis	Reference Documents and Acceptance Norms	Testing Agency	Remarks	Checking Authority
(iii)	Preperation of mix and taking of samples for strength (7days, 14 days & 28 days) and slump report	100%	Technical specification IS 456 and other relevant standards	Contractor/ APTRANSCO approved test laboratory	Counter check of test results and approval by APTRANSCO	AE/AEE/ EE
iv)	Review of controlled concrete mix report	100%	Technical specification IS 456 and other relevant standards	Contractor/ APTRANSCO approved test laboratory	Approval by APTRANSCO	AE/AEE/ EE
a)	Mixing of concrete	100%	Technical specification IS 456 and other relevant standards	Concrete Mixer, water drum, GI sheets, weigh mixing. Arrangement Approved concrete design mix report/ contractor		AE/AEE
(b)	Padding/lean concrete base	100%	DO and approved drawing	DO	DO	AE/AEE
©	Reinforcement placing fixing as per bar bending schedule 2. Check for proper cover spacers and chairs. 3. Check whether lapping of bars are tied properly.	100%	Technical specification IS:2502 and other relevant standards and approved drawing	DO	DO	AE/AEE/ EE
(d)	Fixing of form boxes check for identification, dimensions parallisation and squareness and equidistance from stubs	100%	Technical specification IS 456 and other relevant standards approved drawings	DO	DO	AE/AEE
(e)	Placing of concrete and Ramming	DO	Technical specification IS 456, IS:5613 and other relevant standards	Poking rods and concrete vibrator. Approved concrete design mix report/ Contractor	DO	AE/AEE
(f)	Fixing of chimeny colomn width length check square & parallelism & equidistance from stub	DO	DO	DO	DO	AE/AEE
(g)	Placing concrete, poking & ramming	DO	DO	DO	Gap between boxes & reinforcement rods should be maintained	AE/AEE
F)	**Box removal check for dimensional conformity and workmanship**	100%	Technical specification IS 456, and other relevant standards appoved Drgs.	APTRANSCO & Contractor	Reinforcement rods should not be exposed Boxes should not be opened after 24 hours. Approval by APTransco	AE/AEE

Sl. No.	Component Operation & Description of Test	Sampling Plan with Basis	Reference Documents and Acceptance Norms	Testing Agency	Remarks	Checking Authority
G)	**Backfilling**	100% all Locations	Technical Specification/IS: 5613, and other relevant standards Boulder size 80 mm (Max)	Contractor	Boulders of size more than 80mm should not be allowed for backfilling, sufficient water should be poured and rammed for every 200mm layer of earth, 100mm height earthen bund along with the side of excavation for filling water	AE/AEE/ EE
H)	**Coping and Curing**	100%	Technical Specification IS:5613, and other relevant standards and approved drawing	Contractor	At locations in the paddy fields or in the low lying areas, the chimney is to be extended as per AP Transco Approved drawing. The corner reinforcement rods 200 mm below the final chimney level to be left un concerted and the extensions of chimney for this portion and coping shall be done after tower erection. Curing should be continued for minimum 15 days.	AE/AEE
I)	**Laying of Earthing (Pie or Counterpoise type). (Based on measurement of soil resistivity/ tower footing resistance)**	100% all Locations	Technical specification IS 3403	Joint inspection by APTRANSCO & contractor	Earthing type will be decided by APTRANSCO	AE/AEE
6 A)	**<u>Concrete Cubes Testing</u>**	100% one sample for each location (One sample consists of min 4 test cubes for 28 days strength	Technical specification. IS 456, 516, 1199 and other relevant standards	contractor/. Approved laboratory by APTransco Standard Cube Moulds	Cubes mut be tested within a week of 20 days curing period and tests results should be approved before tower erection by APTRANSCO	AE/AEE/ EE
(i)	Strength under compression	DO	DO	DO	Approval by APTRANSCO	EE/SE

(ii)	Slump test	DO	DO	DOtest Funnel and poking rod	DO	AEE/EE
iii)	Unit weight of concrete	DO	DO	DO	DO	AEE/EE
B)	**Bricks**					
a)	Ansorption Strength	After 24 hours immersion in cold water	IS:3495 - 1966 Not more than 20% by weight	Contractor / approved testing LAB by APTRANSCO		AEE/EE/ SE
b)	Compresive Strength	……..do……..	Not less than 40 Kg/Cm	……..do……..		AEE/EE/ SE
c)	Efflorescence	……..do……..	Not more than "Moderate"	……..do……..		AEE/EE/ SE
C)	**Earth work and compaction control**					
a)	Degree of compaction	1 test per 2000 Sq.m area	IS: 2720 (Par-28)	Contractor / approved testing LAB by APTRANSCO	Review and acceptance of test results by APTRANSCO	SE/EE
D)	**Roads**					
	Water Bound Macadam					
a)	Aggregate impact value	1 test per 250 Cum	IS: 2386 (Part-4)	Contractor / approved testing LAB by APTRANSCO	Contractor / approved testing LAB by APTRANSCO	SE
b)	Grading of aggregate and screening	2 tests per 250 Cum	IS: 2386 (Part-1)	……..do……..	……..do……..	SE
c)	Flakiness index and elongation index	1 test per 250 Cum	IS: 2386 (Part-1)	……..do……..	……..do……..	SE
d)	Atterberg limits of blinding material	1 test per 50 Cum	IS: 2720 (Part-5)	……..do……..	……..do……..	SE
e)	Water absorption	1 test per source	IS: 2386 (Part-30)	……..do……..	……..do……..	SE
f)	Thickness	Regularly	……..	……..do……..	……..do……..	AE/AEE
7	**<u>Tower Erection</u>**					
A)	**Material Checking**					
i)	Checking of tower members for damage cleanliness, galvanizing & stacking	100&	Technical Specifications IS:5613 & other relevant standards and approved drawings	Contractor	Approval by APTRANSCO	AE/AEE/ EE
(ii)	Checking of galvanized bolts & nuts step bolts d'shackle U bolts Spring Washer & accessories	DO	DO	DO	DO	DO
iii)	Dimensional check	Random	DO	DO	DO	DO

B)	**Erection of Super Structure**					
(i)	Checking of leg members/bracing/redundant size & proper placement	Random	Technical Specifications IS:5613 & other relevant standards and approved drawings. Tightening of Bolts & Nuts with torque wrenches	Joint inspection by AP Transco & contractor	Before starting erection of upper section ensure that (i) the lower section is completely braced and all bolts provided in accordance with approved drawing (ii) all plan diagonally and redundants placed in position. Support guys at main panels are used during erection to avoid member deformations. Approval by APTransco.	AE/ AEE
(ii)	Tightness of bolts identification cleanliness & galvanizing	100% each location	Technical IS:5613 and other relevant standards and other approved drawings	Joint inspection by AP Transco & contractor	- do -	AE/ AEE
(iii)	Punching of tightened bolts	- do -	- do-	- do -	- do-	AE/ AEE
(iv)	Checking of assembly & verticality	- do -	- do-	- do -	- do-	Verticality shall not exceed 1: 360

(v)	Tack Welding	100% each location	Technical IS:5613 and other relevant standards and other approved drawings	Joint inspection by AP Transco & contractor	Approval by APTRANSCO	Tack welding dia metrically opposite, length of each welding at least 10mm, 2 coats of Zinc rich paint on welded portion
vi)	Tower footing resistance measurement before & after earthing	- do -	- do -	Measurement by 4 electrode methods by contractor	Approval by APTransco	AE/ AEE
vii)	Fixing of danger plates, number plates & phase plates, circuit plates (i)Visual Check (ii) Dimensional check	- do -	- do -	By contractor	Approval by APTransco	AE/ AEE
8	**Line Stringing**					
A)	Stringing (Sag Tension) chart	100%	Technical Specification IS:5613, Sag Tension calculations and stringing chart approved by APTransco	Contractor	Approval by APTransco	AE/ AEE/ EE
B)	Checking of insulators identification cleanliness packing glazing	100%	Technical Specification IS:731,5613,IEC 383 and other relevant	Joint inspection by APTransco & contractor	Approval not necessary	- do-

	cracks & white spots, security clips		standards approved sag tension calculation and approved stringing chart			
C)	Measurement of IR Value	100%	Technical Specification, IS:5613, and other relevant standards approved insulator megger dbe drawing,	APTransco/ contractor	Approval not necessary	IR shall be more than 50 M Ohms
D)	Checking of insulators hardware fittings and conductor & ground wire accessories	100%	Technical Specification, IS: 2121,2486,5613 and other relevant standards and approved drawings	APTransco/ contractor	Approval not necessary	
(i)	Identification, Cleanliness & packing	- do -	- do-	- do -	- do-	
(ii)	Any mechanical damage	- do -	- do-	- do -	- do-	
(iii)	Measurement of dimensions before and after making joint	- do -	- do-	- do - Hydraulic Compressor, measuring tape and scale vernier caliper/contractor & APTransco	Approval not necessary	
(iv)	Checking mechanical strength	Random	Technical specification IS: 2121,2486,5613 and other relevant standards and approved drawings	A rigging set up to be fabricated in the field for testing mechanical strength. Contractor	Approval not necessary	Mech. Strength not to be less than 95% of UTS of conductor

E)	**Checking of conductor & Earth wire**					
(i)	Drum schedule approval	100%	Technical specification Approved Tower schedules,	Section lengths, Drum invoice lot wise/contractor	Counter check and approval by APTransco	
(ii)	Identification cleanliness & packing	100%	Technical specification and IS:1778,5613 & other relevant standards and other approved drawings	Joint inspection by APTransco & Contractor	Approval by APTransco	
(iii)	Damage of conductor & earthwire stranding	- do -	- do-	- do -	- do-	
(iv)	Drum rubbing against ground & or any metal part	- do -	- do-	- do -	- do-	
F)	**Conductor Earthwire Stringing**	Entire Route	Technical specification IS:5613 & other relevant standards and approved SAG & Tension charts	Joint inspection by APTransco & Contractor	Approval by APTransco	Use of TSE, sagging winch Hydraulic dynamometer and other tools etc.
(i)	Initial conductor position	- do -	- do-	- do -	- do-	
(ii)	Check for Temperature	- do -	- do-	- do -	- do-	Metal enclosed thermometer
(iii)	Final conductor position	- do -	- do-	- do -	- do-	Considering creep
(a)	Dimensional Continuity	- do -	- do-	- do -	- do-	

(b)	Joints in conductor and Earthwire	- do -	- do-	- do -	- do-	No bend curve in the joint
(iv)	Jumpering (Performed)	- do -	- do-	- do -	- do-	Minimum clearance to be maintained as 2.10mt for 220 KV and 1.54 mt for 132 kv
(v)	Tension Stringing (if any). Installation of pilot string on DD Tower	- do -	- do-	- do -	- do-	- do -
9	**Final Checking**					EE/SE
(a)	i) Check for Backfilling ii) Removing any stored material near to the tower iii) Store extra back fill within the base of the tower iv) Checking of prospective revetment with weep holes and store packing of weep holes.	100%	Specifications, IS:5613 and other relevant standards and approved Drgs.	Join inspection by APTransco & contractor	Approval & Acceptance by APTransco	All excess material should be cleared away as per specification
(b)	Check for ACD & all tower accessories	Entire Route	- do -	- do-	- do -	
(c)	Check for Tightening, punching and tack welding of bolts	- do -	- do-	- do -	- do-	

			20			
(d)	Check for Final ground and electrical clearances	- do -	- do-	- do -	- do-	
(e)	Check for effectiveness of Earthing	- do -	- do-	- do -	- do-	
(f)	Meggering of the Line	100%	- do-	- do -	- do-	
(g)	Conductor continuity Test	- do -	- do-	- do -	- do-	
(h)	Measurement of line resistance	- do -	- do-	- do -	- do-	
(i)	Phase sequence	- do -	- do-	- do -	- do-	
10	Testing & Commissioning the Line	100%	Technical Specification	Testing agency APTransco	- do-	
11	Post commissioning checks a) Check for visual corona at insulator, hardware, conductor, accessories etc.(locations to be identified by APTransco)					

Note:

1. Route Plan, approved profiles, approved tower schedule sag template etc will be made available by the purchaser if the detailed survey is already carried out by the Purchaser
2. At the time of receipt of material i.e., Tower parts, Conductors, Hardware fitting accessories, insulators earthwire etc. AP Transco clearance may be verified by contractor.
3. APTRANSCO site office Engineers/Field quality Engineers to witness/review specified tests.
4. The T & P of the erection contractor shall be inspected at site by the APTRANSCO before their use.

GENERAL GUIDE LINES FOR IMPLEMENTATION

Section: GENERAL GUIDELINES FOR IMPLEMENTATION

11 CEMENT

11.1 In case supply of cement is in the scope of the contractor, the same shall be procured from sources approved by APTRANSCO site and got tested at site on sample basis for specified acceptance tests as specified in this FQP at a reputed Third Party Lab approved by APTRANSCO site.

11.2 The samples of cement for site testing shall be taken within three weeks of the delivery and all the tests shall be commenced within one week of sampling.

12. REINFORCEMENT STEEL

12.1 In case supply of reinforcement steel is in the scope of the contractor, the same shall be procured from the main producers i.e., SAIL, TISCO, HSCO or Rashtriya Ispat Nigam or the recollers approved by main procedures. The reinforcement steel shall be got tested at site on sample basis for specified acceptance tests as specified in this FQP at a reputed Third Party Lab approved by APTRANSCO site.

12.2 The result of the testing of cement and reinforcement steel referred to in 13.1 and 14.1 above shall be got approved from APTRANSCO site before cement and reinforcement steel are put to use. However, in exceptional cases due to exigencies of work. APTRASNCO site may authorize the contractor to use Cement and Reinforcement Steel even before the test results are received. However, in all such cases, if the test results subsequently received are found to be not complying with the specified acceptance criteria, the contractor shall have to dismantle and recase all such foundations cast with such non-conforming materials at his own cost. Confirmation to this effect shall be obtained from the contractor by the Project authorities beforehand in all such cases.

13. All the testing and measuring equipments used by the contractor for testing are required to be calibrated. A copy of valid calibration report shall be retained by APTRANSCO as records.

14. Calibration certificates to be got verified from APTRANSCO for tools and plants and testing equipment before their use.

TOWER FOUNDATION

1. CHECK SURVEY

Name of Line: Location No:
Name of Contractor: Type of tower:

i) Approved Detailed Survey Profile Drg. Nos. ……
ii) Reference level ……

iii) Peg Marked Suspension Locations w.r.t Detailed Survey Profile O.K./NOT O.K.
iv) Alignment of Loc. w.r.t previous and next loc O.K./NOT O.K.

v) Loc. Of center peg/position of various landmarks are matching as per profile O.K./NOT O.K.

vi) Any new features on grnd. w.r.t profile necessitating realignment/shifting of loc. Yes/No

		As per profile (Mtrs.)	Actual (Mtrs.)
vii)	Span on both sides of loc.		
	a) Preceding span (Loc.No.)		
	b) Succeeding span (Loc.No.)		
viii)	Angle of deviation and bisection at tower loc.	As per profile	Actual
	a) Angle of deviation		
	b) Bisection		

Certificate:- Cleared for Pit marking.

FOR CONTRACTOR

Signature……

Name……

Designation……

Date……

FOR APTRANSCO

Signature……

Name……

Designation……

Date……

2. PIT MARKING

	Name of Line:	Location No:-
	Name of Contractor:-	Type of tower:-
i)	Approved Pit Marking Drg. Nos.	.....
ii)	Angle of Bisection	O.K./NOT O.K.
iii)	Position of pegs in line/longitudinal and transverse direction	O.K./NOT O.K.
iv)	Position of leg pegs w.r.t center peg of tower	O.K./NOT O.K.
v)	Position of all four pits are on leveled grnd.and safe	Yes/No
vi)	Dimensions of pit marking are per drgs.	O.K./NOT O.K.
vii)	Whether Benching/Revetment required if yes, then	Yes/No
	a) If contour maps/Revetment drgs. Prepared	Yes/No
	b) Calculate vol.	.....cu.m.
	c) Approved by designated authority	Yes/No

Certificate:- Cleared for excavation.

FOR CONTRACTOR

Signature.....

Name.....

Designation.....

Date.....

FOR APTRANSCO

Signature.....

Name.....

Designation.....

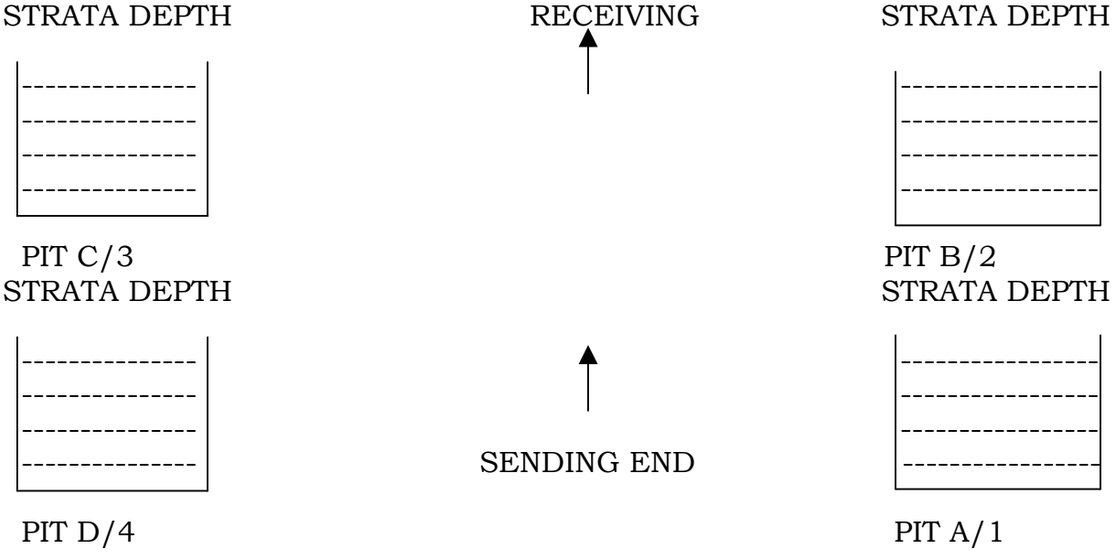
Date.....

FOUNDATION CLASSIFICATION

Name of Line: Location No:-

Name of Contractor:- Type of tower:-

i) Strata wise pit details



NOTE:CLASSIFICATION MAY ALSO BE GIVEN ON ONE PIT EXCAVATION BASIS FOR NORMAL DRY SOIL FOUNDATION DIMENSION

a) Predominant soil
ii) Sub soil water table details as on date.

PIT A/1	PIT B/2	PIT C/3	PIT D/4

iii) Water table in near by well as on date .....mtrs.

iv) Maxm. Subsoil water table in monsoon season after through local enquiry .....mtrs.

v) Surface water table on grnd. In monsoon season and its duration. .....mtrs.for.....days

vi) Type of cultivation Paddy fields/cultivate land/barren land

Whether encasement of stubs reqd Due to surface water? Yes/No

If yes, ht. of encasement above grad.level. .....mtrs.

vii) Whether soil investigation (SPT) carried out at this loc. or near by loc. Loc.No.....

viii) If this loc. strata details are comparable with above soil investigation report or any other loc. of this line. Loc. No.....

ix) Details of soil investigation report:-

a) STRATA DEPTH



b) Subsoil water table. .....mtrs.

c) Ultimate Bearing capacity. .....mtrs.

d) Angle of Repose (Φ) .....degrees

x) Foundation classification proposed by contractor.

As per SPT/ Trial pit	Actual

Remarks/Reason:- .....
.....

FOR CONTRACTOR

Signature.....

Name.....

Designation.....

Date.....

FOR APTRANSCO

Signature.....

Name.....

Designation.....

Date.....

Approval

Site visited on.....and details verified
The classification of foundation is approved as.....

FOR APTRANSCO

Signature.....

Name.....

Designation.....

Date.....

4. STUB SETTING

Name of Line:- Location No:-

Name of Contractor:- Type of tower:-

i) Approved drg.Nos.

a) Stub setting template .....
b) Pit dimension .....
c) Stub assembly .....

ii) Pit Dimensions.

a) Depth of pits From Ref. Level Ground Level

Pit A
Pit B
Pit C
Pit D

b) Pit dimensions are as per approved foundation classification Yes/No

c) Excavated soil is kept 2m away from pit edge. Yes/No

d) Under cutting done in case of partially dry, dry Fissured rocks & mixed dry Yes/No

iii) Alignment of template.

a) Tangent Tower
a.a) In the direction of line Yes/No
b) Angle Tower
b.a) Angle of deviation ....degrees
b.b) Alignment of template on bisection Yes/No

iv) Diagonals of template

AC.....m
BD.....m

Φ=Angle of Deviation.
PQ=Line of Bisection.
(Lines AD and BC are perpendicular to PQ)

STUB SETTING ON BISECTION

v)	Level of template checked by dumpy/water level	Yes/No
vi)	Ht. Of template above ground level.	.....Cms.
vii)	Clearance between lowest part of stub and P.C.C. (not less than 15 cm in dry and 20 cm in wet foundation or as specified in the drg.)	Leg A.....cm. Leg B.....cm. Leg C.....cm. Leg D.....cm.
viii)	Jacks placed on firm ground	Yes/No
ix)	Template supports/jacks positioning causing any danger to collapse of pit.	Yes/No
x)	Green concrete blocks placed below stubs to avoid displacement of stub. (only for Prop setting)	Yes /No
xi)	All members of template are fixed as per drg. and fully tightened	Yes /No
xii)	Checking of stubs	
a)	Stub dimensions are as per type of tower	Yes /No
b)	All plates/cleats fixed with reqd.no.of Nut/bolts and are fully	Yes /No
c)	Is earth flat fixed (for tower earthing)	Yes /No

Certificate: Stub Setting Approved.

FOR CONTRACTOR	**FOR APTRANSCO**
Signature.....	Signature.....
Name.....	Name.....
Designation.....	Designation.....
Date.....	Date.....

Availability check of :

i) leveling instruments
ii) Theodo .....
iii) sieves 0.15, 0.3, 0.6, 1.18, 2.36, 4.75, 10,12,20 mm
iv) screw gauge
v) slump cone
vi) No. of cubes cost

5. CONSTRUCTION OF MATERIALS

Name of Line:- Location No.:-
Name of Contractor:- Type of tower:-

i) Quality /Qty. of Cement as per approved Drgs./Specifications/ Agreement /Booklet on Guidelines on Implementation of QAP

Quantity Required	Quantity Available at Location	Approved Source	Quality (Ok/Not Ok)
			1) Stacking 2) Organic/foreign material 3) Lumps

ii) Quality /Qty. of Sand/Fine Aggregate as per approved Drgs./ Specifications / Agreement / Booklet on Guidelines on Implementation of QAP

Quantity Required	Quantity Available	Approved Source	Quality (Ok/Not Ok)
			1) Stacking 2) Organic/foreign material 3) Lumps 4) Fine Sand Silt and if no 5) Sieves quotable.

iii) Quality/Qty. of Metal/Coarse Aggregate as per approved drgs. And specifications.

Size	Quantity Required	Quantity Avail	Approved Source	Quality (Ok/Not Ok)
40mm(P.C.C) 20 mm (Slabs& Chimney) 10 mm (15 to 20% of 20 mm)				i) Stacking ii) Organic/Profile material iii) Flackiness, soft hammer shaper

iv) Dimensions of the measuring Box. OK/NOT OK
(30 cm x 30 cm x 39 cm Height)

v) Proportions of concrete Mix. by volume derived from controlled design mix by weight.

Grade of Concrete	**Qty. of Coarse Aggregate**	**Qty.of fine Aggregate**	**Qty. of Cement**	**Qty. of Water**
M-15	6 Boxes	3 Boxes	1 Bag	34.1 litres
M-20	4 Boxes	2 Boxes	1 Bag	28 Ltrs.

* Volume proportioning of material by boxes to be verified & conformed after finalisation of Controlled Design Mix by weight.

vi) Quality & Qty. of water as per specns. OK/ NOT OK

vii) Reinforcement steel.
a) Qty. & Quality as per Apprd. Drg./specns.

Dia.(MT)	Qty. Reqd.(MT)	Qty. Avail	Apprd. Source	Quality (Ok /Not Ok)
6 mm 8 mm 12 mm 16 mm 22 mm 32 mm …mm …mm				

viii) From Boxes.
a) Dimensions are as per Apprd. Drgs. OK /NOT OK
b) Oiling of inner wall of form boxes OK /NOT OK

Certificate:- Material & T &P cleared for lean concreting

ix) Lean concreting:-

a) Pits are free from all Foreign Materials Yes/No
b) Pits are free from standing water (Dewatering continued in advance by Pumps / Buckets) Yes/No
c) Mix ratio 1:3:6 with 40 mm metal Yes/No
d) Concrete mixing by mixer as per specn. Yes/No
e) Mixture running time (mixing Time 2 ½ mins.) Yes/No
f) De-watering done. Yes/No
g) Lean concreting done upto specified level and in all the four pits. Yes/No
h) No. of cement bags consumed As per design Actual

x) In case of excess excavation filling is done by lean concrete & no loose soil is permitted for filling.

Vol. of excess lean concrete.

Certificate:- Pits are cleared for installation of reinforcement and form boxes.

FOR CONTRACTOR	**FOR APTRANSCO**
Signature……	Signature……
Name……	Name……
Designation……	Designation……
Date……	Date……

6. INSTALLATION OF REINFORCEMENT STEEL & FORM BOXES

Name of Line:- Location No.:-

Name of Contractor:- Type of tower:-

i) Apprd.Drg.Nos. ----------

---

ii) Reinforcement steel

a)	Quality/Qty. as per specification	Yes/No
b)	Bending /Placing as per approved drg.	Yes/No
c)	Required no. of chairs approved (Min Dia-12 mm, Max. spacing-500mm)	Yes/No
d)	Binding done as per specifications.	Yes/No
e)	Any undue stress or bending of steel bars	Yes/No
f)	Steel is clean and free from loose rust or any other foreign matls	Yes/No
g)	Position of bars w.r.t stub as per drg.	OK /NOT OK

iii) Form Boxes:-

a)	Dimensions as per approved drg.	OK /NOT OK
b)	Placing w.r.t stub as per approved drg.	OK /NOT OK

iv) Clear cover of 50 mm (or as per specification available) Yes/No

Certificate:- Cleared for foundation casting.

FOR CONTRACTOR	**FOR APTRANSCO**
Signature.....	Signature.....
Name.....	Name.....
Designation.....	Designation.....
Date.....	Date.....

7. MIXING, PLACING AND COMPACTION OF CONCRETE

Name of Line:- Location No.:-

Name of Contractor:- Type of tower:-

i)	Apprd.Drg.Nos.	----------
ii)	Mix Ratio	----------
a)	For Pyramid/base with 20 mm metal with specified ratio	Yes/No
b)	For chimney with 20 mm metal with specified ratio	Yes/No
c)	Water to cement ratio as per specification	Yes/No
iii)	Mixing done by Mixer (Running time 2 ½ mins.)	
iv)	Use of poking rod for compacting	Yes/No
v)	Use of vibrator for compacting	Yes/No
vi)	Checking of templates levels & its diagonals at regular intervals	OK /NOT OK
viii)	Casting of legs done in continuity	Yes/No
ix)	No. of cement bags consumed	As per drg. Actual
x)	Removal of stub template, shoring & shuttering and stripping of form work after 24 hours of concreting, observe the damage and make rectification	Yes/No
xi)	Back filling and compaction of backfilled as per specification	Yes/No
xii)	Curing of backfilled earth & chimney carried out for period as per specn. (Minimum period of curing 14 days after concreting)	Yes /No
a)	Availability of water pump with storing arrangement for curing purpose	----------
b)	Date of start of curing	----------
c)	Date of completion of curing	----------
xiii)	Cubes, last sent for testing as per approved FQP	Yes/No
xiv)	All the surplus materials removed from site	Yes/No

Certificate:-Foundation is cleared for tower erection subject to fulfillment of Part (I) before tower erection and Part (II) in due course as follows.

Part-I Setting period (28 days) is allowed as per specification

Part-II (i) Revetment benching proposal status.....

(ii) Revetment benching likely execution date.....

FOR CONTRACTOR

Signature.....

Name.....

Designation.....

Date.....

FOR APTRANSCO

Signature.....

Name.....

Designation.....

Date.....

CONCRETE CUBE TEST REPORT

Name of work :

Agreement No. :

Sample No. :

Identification Mark :

Portion of Work or Qty. represented sample:

Date & Time of casting :

Proportion of Mix :

Duration of Curing at site and
Site temperature before
Transportation to test Lab
and Temperature

		7 days	28 days
1.	Due date of test	.....	.....
2.	Actual date of testing	.....	.....
3.	Delay in testing	.....	.....
4.	Actual Average compressive strength	.....	.....
5.	Is average comp. Strength equal to or more than specified comp.strength	Yes/No	Yes/No

(Signature of Executing Agency) **(Signature of APTransco Official)**

Date..... Date.....

CHECKING OF FOUNDATION MATERIAL

(A) CEMENT

i)	Approved brand of cement (Ramco/Pennar/Raasi)	OK/NOT OK
ii)	Stacking of cement bags	OK/NOT OK
	a) Protective covering found	OK/NOT OK
iii)	Condition of cement bags	Damaged/Undamaged
iv)	No. of cement bags leaking	----------

(Signature of Executing Agency) **(Signature of APTransco Official)**

Date…… Date……

(B) COARSE & FINE AGGREGARE

i)	Stacking of aggregates		OK/ NOT OK
ii)	Size of aggregated Coarse aggregate	20 mm (80 to 85%) 10 mm (15 to 20%)	
	Fine aggregate /sand	4.75 mm conforming to Grade Zone – I sand.	
iii)	Organic material and foreign material and if yes ensure cleaning		Yes/No
iv)	Coarse aggregate-Flake ness, soft slice and laminar shape		Yes/No
v)	Fine aggregate/Sand-Lumps and silt and if yes sieve available for cleaning		Yes/No Yes/No

(Signature of Executing Agency) **(Signature of APTransco Official)**

Date…… Date……

(C) REINFORCEMENT STEEL

i) Approved reinforcement bending schedule Drawing No.....

ii) Dimensions and nos of the reinforcement bars and rigs Yes/No

iii) Cleanliness of reinforcement bars from rust, mill deposit etc.,Yes/No

iv) Binding wire availability Yes/No

v) adequate fabricated chairs and gitties for supporting of reinforce bars availability Yes/No

(Signature of Erection Agency) **(Signature of APTransco Official)**

Date..... Date.....

FORMAT FOR TEST REPORT OF WATER

Name of line……. Source of water supply…….

Sample Indent Mark…….

Name of Executing Agency……. Date…….

Name & address of Test Lab …….

…….

…….

…….

Test Report Ref. No…….

Sl.No.	Description	Requirement/Tested as per	Permissible limits (Max.)
1	Water	Potable	---
2	Naturalization 100 ml H_20	IS: 3025	5ml 0.02 Normal Na oH
	* Using Phenolphthalein as indicator * Using Mixed indicator		25ml 0.02 Normal Na oH
3	PH Value	IS:456	Not less than 6
4	Solids		
	a) Organic	IS:3025	200 mg/l
	b) In Organic	IS:3025	3000 mg/l
	c) Sulphate (SO_3)	IS:3025	4000 mg/l
	d) Chlorides	IS:3025	2000 mg/l (for non-reinforced concrete)
	e) Suspended Matter	IS:3025	500 mg/l (for reinforced concrete)
5	Compressor strength (20 days) for three 150 mm concrete cubes made with proposed water Vis-a Vis similar cubes made with distilled water	IS: 516	90%

6	No. Description	Requirement/ Tested as per	Permissible limits (Max.)
7	a) Initial setting time for test cube made with proposed water	IS: 4031	Not less than 30 months
	b) Variation in initial setting time for test cube made with proposed water and distilled water		30 months

Note: Latest values may be seen by referring the latest IS code as amended as per Relevant standards (IS 3025, 456, 516, 4031) and APTRANSCO Technical Specification.

(SIGNATURE OF LOCAL OFFICIAL)

SIGNATURE OF ERECTION AGENCY

DATE.

DATE……

SAMPLE IS APPROVED FOR CONCRETING

(SIGNATURE OF APTRANSCO OFFICIAL)

DATE……

CEMENT REGISTER

CHECKING OF FOUNDATION MATERIAL

(A) CEMENT

Records of cement testing as
Per approved SFQP

(a) Details of samples
per 200 MT from each source are
maintained as per SFQP ----------

(b) All the tests as per approved SFQP are
carried out and details maintained ----------

(c) Joint reports for collection of samples
in case of site testing are available ----------

(Signature of Erection Agency) **(Signature of APTransco Official)**

Date…… Date……

COARSE/FINE AGGREGATE REGISTER

CEMENT RECEIPT & TEST FREQUENCY RECORD

Name of Line…… Location of ……

Name of Erection Agency…… Source of supply……

Sl.No	Date of Receipt	Batch No./Lot No.	Qty. Received Bags/MT	Cum Qty. Received Bags/MT	Manufacturer's Test Certificate No. & Date	APTRANSCO Approved Lab test Report No. & Date	Test Results are as per Approved SFQP	Signature

Note: (1) MTC shall be available for every 200 MT or part thereof
(2) Lab Test Report shall be available for every 200 MT or part thereof

FORMAT FOR JOINT SAMPLING OF CEMENT

We have collected the sample of Cement as per following details joined:

1. Location of stores it: _____

2. Source of supply : _____

3. Type and Grade of cement: (OPC/PPC, grade 33.43.53) _____

4. Batch No. /Lot No. of cement _____

5. No. & Quantity of sample _____

6. Name & Address of Lab where samples are being sent for testing (vide letter No. & date) _____

The samples have been sealed and one identification mark slip duly signed her been placed inside the bag & another slip duly signed has been placed on the bag.

(Signature of Erection Agency) Signature of APTransco Official

Date…… Date……

FORMAT FOR TEST REPORT OF CEMENT

Name of line _____Source of Supply_____
Name of Executing Agency _____Sample Ident. Mark_____
Test Report Ref. No._____ Date:_____
Name & Address of Test Lab_____

Sl.No	Particulars	Requirement as per				As per 3rd party Approved Lab. Test Result
		IS 269 33 Grade OPC	IS 8112 43 Grade OPC	IS 12269 53 Grade OPC	IS 1489-I & II, Fly Ash or Clay based	
A.	Chemical Properties				--	
1.	Cao-0.7 SO3	0.66 to 1.02	0.66 to 1.02	0.8 to 1.02		
2.	$2.8SiO_2+1.2A1_2O_3+0.65Fc_2O_3$					
3.	20_3% $Fc20_3$	0.66 MIN	0.66 MIN	0.66 MIN	--	
4.	Insoluble Residue (%by mass)	4.0 MAX	2.0 MAX	2.0 MAX	13.6 to 2.8**	
5.	Magnesia(%by mass)	6.0 MAX	6.0 MAX	6.0 MAX	6.0 MAX	
6.	Sulphuric Anhydrite(% by mass)	3.0 MAX	3.0 MAX	3.0 MAX	3.0 MAX	
7.	Total loss of ignition(%)	5.0 MAX	5.0 MAX	5.0 MAX	5.0 MAX	
8.	Chloride(%)	0.05 MAX	0.05 MAX	0.05 MAX	--	
B.	Physical Properties					
1.	Fineness (SQM/Kg.)	225 MIN	225 MIN	225 MIN	300 MIN	
2.	Setting time (minutes)					
	i) initial	30 MIN	30 MIN	30 MIN	30 MIN	
	ii) final	600 MAX	600 MAX	600 MAX	600 MAX	
3.	Soundness					
a)	Le-chat Expansion(MM)					
	i) Unacrated cement	10.0 MAX	10.0 MAX	10.0 MAX	10.0 MAX	
	ii) After 7 days Acration	5.0 MAX	5.0 MAX	5.0 MAX	5.0 MAX	
b)	Auto clave expansion (%)					
	i) Uncrated cement	0.8 MAX	0.8 MAX	0.8 MAX	0.8 MAX	
	ii) After 7 days Aeration	0.6 MAX	0.6 MAX	0.6 MAX	0.6 MAX	
4.	Compressive Strength (Mpa)					
	a) 72+/- 1 hour	16 MIN	22 MIN	27 MIN	16 MIN	
	b) 168 +/-2 hours	22 MIN	33 MIN	37 MIN	22 MIN	
	c) 672+ /-4 hours	33 MIN	43 MIN	53 MIN	33 MIN	

** X+4.0(100-X)/100 Where X is the declared 1% of pozzolona in the given Portland pozzolona cement. X Ranger from 10 to 25%.

Note: Latest values may be seen by referring the latest IS code as amended.

COARSE & FINE AGGREGATE

Record of Coarse Aggregates testing as per approved SFQP

a) Details of samples per 200 cum from Each source for each size (40mm, 20mm) And fine aggregate are maintained as per SFQP ----------

b) All tests as per approved SFQP are carried out and details maintenanced ----------

c) Joint Report for collection of Samples for testing at APTRANSCO Approved lab is available ----------

(Signature of Erection Agency)

Date……

Signature of APTransco Official

Date……

COARSE & FINE AGGREGATE CONSUMED & TEST FREQUENCY

.....

Name of Line ---------- 40 mm metal----------
20 mm metal

Name of Erection Agency ----------

Sl No	Date of Casting	Loc No.	Type of Foundation	40 mm metal		20 mm metal		Sand		Signature	
				Consumed	Commutative MT	Consumed	Commutative MT	Consumed	Commutative MT	Erection Agency	APTRANSCO

Note: One sample per lot of 200 MT or part there of (each for 40 mm, 20 mm metal and sand).....Date of sampling and test report reference to be recorded in this format.

FORMAT FOR JOINT SAMPLING OF COARSE & FINE AGGREGATE

The sample of coarse line aggregate as per the following details have been collect …….

(1) Location of sit

(2) Source of supply

Size of Aggregate	Source
40 mm	…….
20 mm	…….
Fine Aggregate	…….

(3) No. & Quantity of Samples taken

(4) Name & address of
Lab where samples are
Being sent for testing
(vide letter No. & date)

The samples have been sealed and identification mark slip duly signed has been placed inside the bag & another slip duly signed has been pasted on the bags.

(Signature of Erection Agency) Signature of APTransco Official

Date……. Date…….

FORMAT FOR TEST REPORT OF FINE AGGREGATES

Name of Line_____Source of Supply_____

Name of Executing Agency_____Sample Ident.Mark_____

Name & address of Test lab_____Date_____

Test Report Ref.No._____

Sl No	Description of test	Acceptable Limits As per IS 383				Test Results	Remarks
		Percentage of Passing Zone wise					
		Zone-I	Zone-II	Zone-III	Zone-IV		
1.	Sieve Analysis						
	10 mm	100	100	100	100		
	4.75 mm	90-100	90-100	90-100	95-100		
	2.36 mm	60-95	75-100	85-100	95-100		
	1.18 mm	30-70	55-90	75-100	90-100		
	600 micron	15-34	35-59	60-79	80-100		
	300 micron	5-20	8-30	12-40	15-50		
	150 micron	0-10	0-10	0-10	0-15		
2.	Silt content	8% Max.	8% Max.	8% Max.	8% Max.		
3.	Deleterious Materials	5% Max.	5% Max.	5% Max.	5% Max.		

Note: Latest values may be seen by referring the latest IS code as amended.

Certificate: It is certified that test results meet all the requirements as per IS383 and APTRANSCO Technical specifications.

(Signature of the Lab Official) (Signature of Erection Agency)

Date:..... Date.....

Sample is approved for concreting/masonary mortar/plastering

(Signature of Powergrid Official)

Date:.....

REINFORCEMENT STEEL REGISTER

REINFORCEMENT STEEL

Record of reinforcement steel
As per approved SFQP.

(a) Details of sample: from each source for each size are maintained as per approved SFQP

(b) All tests as per approved SFQP are carried out and details maintained

(c) Joint Reports for collection of samples for testing at powergrid approved lab is available.

(Signature of Erection Agency)

Date.....

Signature of APTransco Official

Date.....

REINFORCEMENT STEEL RECEIPT & TEST FREQUENCY RECORD

Name of Line……. Location of store…….

Name of Erection Agency……. Source of supply…….

S.No.	Date of Receipt	Grade/type & Lot No.	Diameter	Quantity received		Manufacturer's Test Certificate No. & date	APTRANSCO Approved Lab test Report No. & date	Test Results are As per Approved SFQP	Signature
				Qty. Regd.in Lot (MT)	Cum Qty.MT				

Note: MTC and APTRANSCO approved lab test report shall be available as per approved SMQP.

FORMAT FOR JOINT SAMPLING OF REINFORCEMENT STEEL

The samples of reinforcement steel as per the following details have been collected jointly by us:

(1) Location of store/site

(2) Source of Supply

(3) Diameters of rods

(4) Length of rods

(5) Lot No.(s)

(6) No. of samples taken

(7) Name & Address of lab where samples are being sent for testing (vide letter No. & date)

The samples have been wrapped together & placed inside the bag. One identification mark slip duly signed the bag & another slip duly signed has been pasted on the bag.

(Signature of Erection Agency)

Date.....

Signature of APTransco Official

Date.....

FORMAT FOR TEST REPORT OF REINGORCEMENT STEEL

Name of Line_____Source of Supply_____

Name of Executing Agency_____Sample Ident.Mark_____

Name & address of Test lab_____Date_____

Test Report Ref.No._____

Sl No	Description of test	Acceptable Limits As per			Test Results		Remarks
		IS 432 MS Grade I Above 20 mm dia	IS 1786 Fc 415	Manufac. Specn. TMT 415			
1.	Chemical Analysis	0.23+/-0.02 Max.	0.30+/-0.02 Max	0.25 +/-0.02 Max			
	a) Carbon (C) %	0.055+/-0.005 Max	0.06 +/-0.005 Max	0.05+/-0.005 Max			
	b) Sulphur (S) %	0.055+/-0.005 Max	0.06 +/-0.005 Max	0.05+/-0.005 Max			
	c) Phosphorous(P) %	0.055+/-0.005 Max	0.06+/-0.005 Max	0.05+/- 0.005 Max			
	d) Sulphur and % Phosphorous	0.11+/-0.01 Max	0.11+/-0.01 Max	0.10 +/- 0.01 Max			
2.	Mechanical Properties						
	a) 0.2% proof stress N/SQ.MM	250 Min	415 Min.	415 Min			
	b) Ultimate Tensile Stress N/SQ MM	410 Min	485 Min	500 Min			
	c) Elongation	23 Min	14.5 Min	22 Min			
	d) Bend Test	No crack/Fracture at Bend visible	No crack/Fracture at Bend visible	No crack/Fracture at Bend visible			
	e) Rebend Test	No crack/Fracture at Bend visible	No crack/Fracture at Bend visible	No crack/Fracture at Bend visible			

Note: Latest value may be seen by referring the latest IS code as amended.

Certificate: It is certified that test results meet all the requirements as per IS and APTRANSCO Technical specifications.

(Signature of the Lab Official)

Date:.....

(Signature of Erection Agency)

Date.....

Sample is approved

(Signature of Powergrid Official)
Date.....

CONCRETE CUBE TESTS REGISTER

CONCRETE CUBE TEST REPORT

Name of work :

Agreement No. :

Sample No. :

Identification Mark :

Portion of Work or Qty represented sample:

Date & Time of casting :

Proportion of Mix :

Duration of Curing at site and
Site temperature before
Transportation to test Lab
And Temperature

		7 days	28 days
1.	Due date of test	.....	
2.	Actual date of testing	.....	
3.	Delay in testing	.....	
4.	Actual Average compressive strength	.....	
5.	Is average comp. Strength equal or more than specified comp. Strength	Yes/No	Yes/No

(Signature of Executing Agency) **(Signature of APTransco Official)**

Date..... **Date.....**

TOWER ERECTION

Name of the Line:- Location No:-

Name of Contractor:- Type of tower:-

Approved Drg.No:-

Approved Bill of Material:-

	Items Checked	Results	Observations
1.	Setting period of foundation is allowed for atleast 14 days (preferably 28 days) as per specification	Yes/No	
2.	All tested tools and plants and safety equipments in working conditions are available at site	Yes/No	
3.	All tower members, Nuts,/bolts are available at site without any damage, bent or rusting and kept on ground serially according to erection sequence	Yes/No	
4.	Benching /revetment, if nay, completed. If not, then program of completion	Yes/No	
5.	Shutdown of power line, if required, is arranged	Yes/No	
6.	Required no.of safety helmets, safety belts & safety shoes are being used.	Yes/No	
7.	First section is completely braced and all plane diagonals are place in proper position	Yes/No	
8.	Guying of tower provided as per approved drawings and norms. Guying to be terminated on firm ground.	Yes/No	
9.	All nuts/bolts, flat/spring washers are provided as per approved drawings	Yes/No	
10.	All bolts to have the nuts facing outside for horizontal or near horizontal bolt connections and downwards for vertical bolt connections (Cl.2.27.5.1.1 of the Technical Specification)	Yes/No	
11.	All the inner contact surfaces at joints painted with zinc rice paint (Cl.2.28.3 of the Technical Specification)	Yes/No	

12.	Subsequent section are erected only after complete erection and bracing of previous section	Yes/No
13.	Any undue stress, bending or damage of member during erection noticed	Yes/No
14.	Filling of holes or cutting of members during erection observed	Yes/No
15.	Any heavy hammering of bolt causing damage of threads noticed	Yes/No
16.	Any substitute of tower member erected, if, yes, member nos.	Yes/No
17.	Tightening is done progressively from top to bottom	Yes/No
18.	All bolts at same level on all the four faces tightened simultaneously staring from leg to face on right side by four individuals	Yes/No
19.	Slipping/running over nuts/bolts are replaced by new ones	Yes/No
20.	Threaded portion of bolts projected outside of nut is not less than 3 mm and not more than 8 mm	Yes/No
21.	Punching of threads projected outside is done at three positions on dia	Yes/No
22.	Half round Tack welding of bolts nuts at two diametrically opposite places upto the bottom cross arm and each weld length at least 10mm	Yes/No
22 a)	Zinc rich (at least 90% zinc content) on the tack welds on the nuts	Yes/No
23.	Verticality of tower (1 in 360) is checked with help of Theodolite for both longitudinal & transverse direction. This is within specified limits	Yes/No
24.	Details of missing member, nut, bolts, etc.	Yes/No
25.	Mention the tower taking resistance value:	Yes/No

Certificate: Tower erection is complete in all respects and footing resistance is within permissible limit.

FOR CONTRACTOR
Signature.....

Name.....

Designation.....
Date.....

FOR APTRANSCO
Signature.....

Name.....

Designation.....
Date.....

TOWER ACCESSORIES

Name of the Line:- Location No:-

Name of Contractor:- Type of tower:-

Approved Drg.No:-

	Items Checked	Results	Observations
1.	GI bolts and nuts fully tightened and placed as per approved drawing.		
	a) Number Plate	Yes/No	
	b) Phase Plates(set of 3 per circuit)	Yes/No	
	c) Danger Plates	Yes/No	
	d) Circuit Plate (2 Nos.)	Yes/No	
	e) Name plate	Yes/No	
	f) Bird guards (Suspension tower only)	Yes/No	
	g) Earthing bond (One on suspension tower and two on tension towers) (37/7/0.417 mm Stranded Tinned Copper)	Yes/No	
	h) Step bolts with spring washers 450 mm spaced on diagonally opposite leg provided 3.5m above the ground level to top of the tower	Yes/No	
	i) Step bolt holes available upto 3.5m above ground level.	Yes/No	
2.	Anti-climbing device including barbed wire, cleats, Barbed wire fixing Galvanized MS member with fully tightened bolts.	Yes/No	
3.	Aviation signals/Paints as per requirement/specification provided where required.	Yes/No	

Certificate: All Tower accessories provided satisfactorily

FOR CONTRACTOR **FOR APTRANSCO**

Signature…… Signature……

Name…… Name……

Designation…… Designation……

Date…… Date……

PIPE TYPE EARTHING

Name of the Line:- Location No:-

Name of Contractor:- Type of tower:-

Approved Drg.No:-

	Items Checked	Results	Observations
1)	Earthing on leg A/1	Yes/No	
2)	GI Pipe (25mm dia, 3m long) Flat (50x6 mm, 5m Long) with nuts and bolts fully tightened and placed as per approved drawing.	Yes/No	
3)	Placement of Flat along with stub and bending of flat inside the form box as per approved drg.	Yes/No	
4)	Burying of Flat 650mm below ground	Yes/No	
5)	Bore hole (300 mm dia) and depth (3000 mm)	Yes/No	
6)	Gap (150 mm) between the GI pipe bottom end and bottom of the bore hole	Yes/No	
7)	Finely broken coke (grain size not more than 25 mm) and salt (proportion 10:1) filled in bore hole around the pipe.	Yes/No	
8)	Mix of Bentonite powder in 1:6 ratio	Yes/No	
9)	Back filling done	Yes/No	
10)	Earthing resistance measurement (less than 10 Ohm) and if not so, then	Yes/No	
11)	Additional earthing pipe installed.	Yes/No	
12)	Repeat earthing resistance measurement (less than 10 ohm)	Yes/No	

Certificate: Footing resistance within permissible limits.

FOR CONTRACTOR **FOR APTRANSCO**

Signature…… Signature……

Name…… Name……

Designation…… Designation……

Date…… Date……

COUNTER POISE EARTHING

Name of the Line:- Location No:-

Name of Contractor:- Type of tower:-

Approved Drg.No:-

	Items Checked	Results	Observations
1)	Earthing provided on all the four legs	Yes/No	
2)	GSS earth wire 7/3.15, 25m long), MS rod (20 mm dia, 6mtrs long) with nuts and bolts fully tightened and placed as per approved drg.	Yes/No	
3)	Burying of GSS earthwire 1000 mm below ground	OK/NOT OK	
4)	MS rod (20mm dia, 600 mm length) at the end of earthwire.	Yes/No	
5)	Back filling done	Yes/No	
6)	Earthing resistance measurement (not less than 10 ohm)	Yes/No	
7)	Additional counter poise installed	Yes/No	
8)	Repeat earthing resistance measurement (less than 10 ohm)	Yes/No	

Certificate: Footing resistance within permissible limits.

FOR CONTRACTOR **FOR APTRANSCO**

Signature…… Signature……

Name…… Name……

Designation…… Designation……

Date…… Date……

STRINGING

Name of the Line:- section: Loc.No……to Loc. No:-

Name of Contractor:- Type of tower:-

Approved Tension Stringing Chart:-

Approved Drum Schedule:-

	Items Checked	Results	Observations
1)	Back filling of soil and revetment/ benching wherever required is done.	Yes/No	
2)	Towers are tightened properly and all the members, nut/bolts are provided.	Yes/No	
3)	Tress in the ROW removed to facilitate smooth stringing	Yes/No	
4)	All line materials, tested and calibrated T &P safety equipments and relevant drawings available for stringing.	Yes/No	
5)	Shutdown of Power line/Railway block if required, is arranged.	Yes/No	
6)	Necessary Protection/scaffolding / warning signals provided for Railway/ Power line/P&T line/Road crossing.	Yes/No	
7)	Towers vulnerable for one side load is guyed properly.	Yes/No	
8)	Tower footing resistance is within permissible limit of 10 Ohms.	Yes/No	

B) Paving Out of Earthwire.

1)	Work is being carried out with fully safety measures.	Yes/No
2)	All pulleys fixed on towers for paving out are moving freely.	Yes/No
3)	Effective communication exists through walkie-talkie and through persons with red flags on towers.	Yes/No
4)	Earthwire is being constantly checked as it is unwound, Damaged portion, if any, is removed.	Yes/No
5)	Necessary arrangement have been provided to avoid rubbing of earthwire against hard ground.	Yes/No
6)	Necessary details of earthwire, M.S.Joints Recorded as per Annex CFI & CF II	Yes/No

C) Paving Out of Conductor

1)	Work is being carried out with full safety measures	Yes/No
2)	Conductor drums are placed properly to avoid bird caging.	Yes/No
3)	Sequence of paving out is such that to avoid unbalancing of load on tower	Yes/No
4)	Details of insulators and fitting are recorded. As per Annex CF III and Annex CF IV.	Yes/No
5)	Travelers fixed on towers are moving freely	Yes/No
6)	Effective communication exists through Walkie-Talkie and through persons with red flags on towers.	Yes/No
7)	Conductors is checked continuously as it is unwound from drum. Damaged portion if any, is removed/repaired.	Yes/No
8)	Travelers fixed on towers are moving freely	Yes/No
9)	Effective communication exists through walkie-talkie and through persons with red flags on towers.	Yes/No
10)	Conductors is checked continuously as it is unwound from drum. Damaged portion if any, is removed/repaired.	Yes/No

11)	Proper arrangements made to avoid rubbing of conductor on ground/hard surfaces.	Yes/No
12)	Details of conductor M.S.J. /repair sleeve are recorded as per Annex CF II and Annex CF II.	Yes/No

D) Final Sagging & Tensioning of Earthwire & Conductor

1)	Sag Board is fixed correctly after taking into account length of suspension clamp/fittings in case of Earthwire and Suspension insulator string in case of conductor.	Yes/No
2)	Sag/Tension is measured correctly at prevailing temperature. Details recorded as per Annex CF V.	Yes/No
3)	Minimum Ground Clearance, clearance over Power line/Railway Line/River crossing/Telecom/Road are as per specification.	Yes/No

E) Dead Ending

1)	Subsequent to final sagging, dead end work is done marking the length of conductor corresponding to single/double tension string for conductor and dead end clamp for ground wire.	Yes/No
2)	Corresponding to dead end clamp drawing, the aluminum strands over the steel core are cut and steel portion and aluminum portion of dead end clamp is compressed as applicable to conductor and only steel clamps for ground wire.	Yes/No
3)	Clipping offsets marked with felt tip marker and not with sharp instruments denting conductor stands.	Yes/No

4) For power line crossing section precautions taken before clipping. Yes/No

i) The existing power line conductors are grounded.

ii) No person in series with the potential electric circuit.

5) Prior to installation/removing dead end jumper on structure earthing done on both sides of the work and connected to structure. Yes/No

6) Suspension clamps and dampers installed with in 36 hours after final connected to structure Yes/No

7) Center of the body (cushioned) and clamp body of AGS clamp coincides with the armor rods, marked point on the conductor. Yes/No

8) No difference in length between the longest and shortest rod when applied not to exceed 13 mm. Yes/No

F) Conductor Fittings and Attachments

1) Spacers and dampers fixed immediately but with in 24 hours of conductor clamping as per manufactures recommendations. Yes/No

2)a) Whether spacer cars/cycles are fixed with meter counters for measuring intra-spacing of spacers/ spacer dampers during placement and if no. Yes/No

b) How the distance between spacers/spacer damper in measured. Yes/No

3) Vibration dampers for ground wire installed as per manufactures recommendations immediately but within 24 hours of clipping. Yes/No

4) Span marker on ground wire, if required, installed on ground wire after clamping and fixing of vibration dampers on the ground wire. Yes/No

G) Tower Hardware

1) Tower Hardware viz phase plates, number plates, circuit plates, copper earthing bond, phase plates and danger plate fixed as per approved drawing. Yes/No

2) Anti-climbing devices fixed as per approved drg. Yes/No

3) Step bolts with 2 Nos. nuts and one spring washer fixed as per approved drawing. Yes/No

Certificate: Stringing is completed in all respects.

FOR CONTRACTOR	**FOR APTRANSCO**
Signature.....	Signature.....
Name.....	Name.....
Designation.....	Designation.....
Date.....	Date.....

ANNEXURE CF I

DETAILS OF EARTHWIRE/CONDUCTOR

1) Make

2) Batch No.

3) Quantity and Location as per approved drum schedule.

Sl. No	Drum No.	Length Marked on Drum (Km)	Length Pay Out (Km)	Pay Out Between		Phase	Wire No.
				From Loc.No.	To Loc. No.		

4) There is no damage to earthwire/conductor before or during stringing standards are in perfect position.

FOR CONTRACTOR

Signature……

Name……

Designation……

Date……

FOR APTRANSCO

Signature……

Name……

Designation……

Date……

ANNEXURE CF II

DETAILS OF COMPRESSION TYPE MID SPAN JOINT FOR EARTHWIRE/ CONDUCTOR AND REPAIR SLEEVE FOR CONDUCTOR

1) Make

2) Batch No.

3) Quantity and Location as per approved drum schedule.

Sl. No	Between Loc No.	Phase	Wire No.

4) Dimension Recorded as per Annex.....

5) M.S. Joint has been provided at least 50 feet away from tower.

6) There is no M.S. Joint over Railway/River/Main Road crossing.

7) Not more than one M.S.Joint provided in one span for each earthwire/conductor

8) Repair sleeve shall be used if number of damaged strands is not more than 2 strands in outer layer of the conductor. No repair sleeve shall be used in case of ground wire. If damage is in more than the permissible strands, the damaged portion shall be removed and M.S. Joint provided.

FOR CONTRACTOR

Signature.....

Name.....

Designation.....

Date.....

FOR APTRANSCO

Signature.....

Name.....

Designation.....

Date.....

ANNEXURE CF III

RECORDS OF INSULATORS

1) Type Glass/porcelain, Suspension /Tension/Pilot.

2) Make

3) Batch No. and Month of Manufacture.

4) With/without zinc sleeve.

5) Electro Mechanical Strength.

6) Quantity and Location

Sl. No.	Loc. No.	Qty. as per specification	Qty. in Ckt I			Qty. in Ckt II			Ckt I			Ckt II		
			Phase (Conductor)			Phase (Conductor)			Jumper			Jumper		
			R	Y	B	R	Y	B	R	Y	B	R	Y	B

7) Insulator Meggered at site with 5 KV Megger and results of measurement.

8) Insulators are cleaned with soft cloth. Glazaing is proper. There is no crack, scratch or white spot on its surface.

9) 'R' clips in insulators are fitted properly.

10) While hoisting, no damage caused to insulators.

FOR CONTRACTOR

Signature……

Name……

Designation……

Date……

FOR APTRANSCO

Signature……

Name……

Designation……

Date……

ANNEXURE CF IV

DETAILS OF HARDWARE FITTING

1) Make

2) Batch No.

3) Type of fitting-I/V Single/Double, Suspension/Tension.

4) Quantity and Location

Sl. No.	Loc. No.	No. of Fittings						Ckt I			Ckt II		
		Phase Ckt. I			Phase Ckt II			Jumper			Jumper		
		R	Y	B	R	Y	B	R	Y	B	R	Y	B

5) All nuts/bolts properly tightened. Yes/No

6) All Components of fittings have been provided as per approved drawings. Dimensions and galvanizing are OK. Fitting is cleaned and there is no damage to any component. Yes/No

7) All spilt pins properly provided. Yes/No

8) In case of Tension fittings, dimensions before and after compression recorded as per Annex……. Yes/No

FOR CONTRACTOR

Signature……

Name……

Designation……

Date……

FOR APTRANSCO

Signature……

Name……

Designation……

Date……

ANNEXURE CF V

SAG MEASUREMENT FOR CONDUCTOR

1) Sag Board/dynamometer fixed between Loc No. and

2) Temperature .....^{o}C

3) Measurement of Sag/Tension.

Item	Phase/Wire No.	As per Sag/Tension Chart		Actual Sag /Tension	
		Initial Sagging	Final Sagging	Initial Sagging	Final Sagging
Sag Tension					

4) During paying out/rough sagging, tension in conductor was as per technical specifications (50% of sagging tension).

5) Calibration certification for dynamometer checked. Yes/No

6) Vertical clearance between top conductor and earth wire at minimum clearance point along the span is indicated as----------

FOR CONTRACTOR

Signature.....

Name.....

Designation.....

Date.....

FOR APTRANSCO

Signature.....

Name.....

Designation.....

Date.....

ANNEXURE CF VI

SUSPENSION/TENSION CLAMPS FOR EARTHWIRE

1) Make

2) Batch No.

3) Quantity and Location

Sl. No.	Wire No.	Loc.No.	Remarks

4) All components of clamps have been provided as per approved drawings. Dimensions and galvanizing are OK. Clamp is cleaned and there is not damage to any component.

5) All Nuts & bolts have been properly tightened, to 5 Kg m torque.

6) Split pins have been properly fixed.

7) In case of tension clamp, dimensions before and after compression recorded as per Annex.....

FOR CONTRACTOR

Signature.....

Name.....

Designation.....

Date.....

FOR APTRANSCO

Signature.....

Name.....

Designation.....

Date.....

ANNEXURE CF VII

RECORDS OF V.D. FOR EARTHWIRE AND CONDUCTOR

1) Make

2) Batch No.

3) Quantity and Location

Sl. No.	Fixed on Loc.No. (Right side)	Fixed on Loc.No. (Left side)	Phase/Wire No.	No. of V.D.

4) All components of V.D. have been provided as per approved drawings. Dimensions and galvanizing are OK. Clamp is cleaned and there is not damage to any component including messenger wire.

5) All Nuts & bolts have been properly tightened, to 4 Kg m torque.

6) V.D. Fixed as per approved placement chart.

FOR CONTRACTOR

Signature……

Name……

Designation……

Date……

FOR APTRANSCO

Signature……

Name……

Designation……

Date……

ANNEXURE CF VIII

RECORDS OF LINE SPACER/SPACER DAMPER/JUMPER SPACER

1) Make

2) Batch No.

3) Quantity and Location

Sl. No.	Span/Loc.No.	Phase	No.of spacer

4) All components of V.D. have been provided as per approved drawings. Dimensions and galvanizing are OK. Clamp is cleaned and there is not damage to any component including messenger wire.

5) All Nuts & bolts have been properly tightened, to 4 Kg m torque.

6) Spacer/Spacer damper Fixed as per approved placement chart. (Copy enclosed.)

FOR CONTRACTOR

Signature.....

Name.....

Designation.....

Date.....

FOR APTRANSCO

Signature.....

Name.....

Designation.....

Date.....

ANNEXURE CF IX
(1)

The Dimensions of Tension Clamp suitable for (Zebra) Moose (AAAC Panther, ACSR Panther, AAAC Moose)

1) Loc.No.

2) Span.....Loc Nos.....to.....

3) Approved Drg.No. EMI/DE/710LT4-31.77

4) Details of Dimensions

Steel Portion						Aluminum Portion									
			As per Drg.			Actual					As per Drg.			Actual		
Before Comp.			After Comp.			After Comp.			Before Comp		After Comp.			After Comp.		
Length		Outer dia		C-C	F-F	Length	C-C	F-F	Length	Outer dia	Length	C-C	F-F	Length	C-C	F-F
Dead End	300 mm	21 ± 1	--	20.2± 0.5	17.5 ± 5				610	54± 1	640	53± 0.5	46± 0.5			
Jumper Leg									230	54± 0.5	--	53± 0.5	46± 0.5			

C-C : Corner to Corner Distance
F-F : Face to Face Distance
For Jumper leg the dimensions before and after compression may be obtained from the supplier and checked.

5) Bores in the sleeves are perfectly clean.

6) The following may be checked as per approved drawing.

   a) Marking and cutting
   b) Correct sizes of dies
   c) Centering & fixing of sleeves
   d) Fixing of all the components i.e., Aluminium end pipes, hole plugs etc.
   e) Compression of sleeves at specified pressure.

7) All the sharp edges have been filled after compression.

8) There is no crack, bend or any damage to joint after compression.

FOR CONTRACTOR

Signature.....

Name.....

Designation.....

Date.....

FOR APTRANSCO

Signature.....

Name.....

Designation.....

Date.....

ANNEXURE CF IX
(2)

Dimensions for 7/3.66 mm GSS Earthwire Tension Clamp

1) Loc.No.

2) Span.....Loc Nos.....to.....

3) Approved Drg.No. EMI/EWTA-27510 0-98
4) Details of Dimensions

Steel Portion						Aluminum Portion									
			As per Drg.			Actual					As per Drg.			Actual		
Before Comp.			After Comp.			After Comp.			Before Comp		After Comp.			After Comp.		
Length		Outer dia	Length	C-C	F-F	Length	C-C	F-F	Length	Outer dia	Length	C-C	F-F	Length	C-C	F-F
Tension Clamps	225 mm	21 ± 0.5	--	20.2 ± 0.5	17.5 ± 0.5				225	32± 0.5	250	29.4 ± 0.5	25± 0.5			
Jumper Leg	80 mm	21 ± 0.5		20.2± 0.5	17.5 ± 0.5				80	32± 0.5	--	29.4 ± 0.5	25± 0.5			

C-C : Corner to Corner Distance
F-F : Face to Face Distance
For Jumper leg the dimensions before and after compression may be obtained from the supplier and checked.

5) Bores in the sleeves are perfectly clean.

6) The following may be checked as per approved drawing.

   a) Marking and cutting
   b) Correct sizes of dies
   c) Centering & fixing of sleeves
   d) Fixing of all the components i.e., Aluminium end pipes, hole plugs etc.
   e) Compression of sleeves at specified pressure.

7) All the sharp edges have been filled after compression.

8) There is no crack, bend or any damage to joint after compression.

FOR CONTRACTOR

Signature.....

Name.....

Designation.....

Date.....

FOR APTRANSCO

Signature.....

Name.....

Designation.....

Date.....

ANNEXURE CF IX
(3)

The Dimensions of Mid Span Joints for Moose Conductors

1) Loc.No.

2) Span……Loc Nos……to…….

3) Approved Drg.No. EMI/MCJC – 73531.77
4) Details of Dimensions

Steel Portion						Aluminum Portion									
			As per Drg.			Actual					As per Drg.			Actual		
Before Comp.			After Comp.			After Comp.			Before Comp		After Comp.			After Comp.		
Length		Outer dia	Length	C-C	F-F	Length	C-C	F-F	Length	Outer dia	Length	C-C	F-F	Length	C-C	F-F
250 ± 5		21 ± 1	285 (approx)	20.2 ± 0.5	17.5 ± 0.5				735 ± 5	54 ± 1	785 (approx)	53 ± 0.5	46 ± 0.5			

C-C : Corner to Corner Distance
F-F : Face to Face Distance
For Jumper leg the dimensions before and after compression may be obtained from the supplier and checked.

5) Bores in the sleeves are perfectly clean.

6) The following may be checked as per approved drawing.

   a) Marking and cutting
   b) Correct sizes of dies
   c) Centering & fixing of sleeves
   d) Fixing of all the components i.e., Aluminium end pipes, hole plugs etc.
   e) Compression of sleeves at specified pressure.

7) All the sharp edges have been filled after compression.

8) There is no crack, bend or any damage to joint after compression.

FOR CONTRACTOR

Signature.....

Name.....

Designation.....

Date.....

FOR APTRANSCO

Signature.....

Name.....

Designation.....

Date.....

ANNEXURE CF IX
(4)

Dimensions for Repair sleeves for ACSR Moose Conductor

1) Loc.No.

2) Span……Loc Nos……to…….

3) Approved Drg.No. EMI/RS – 30531.77

4) Details of Dimensions

		As per Drg.			Actual		
Before Comp.		After Comp.			After Comp.		
Length	Outer dia	Length	C-C	F-F	Length	C-C	F-F

C-C: Corner to Corner Distance
F-F: Face to Face Distance

5) Bores in the sleeves are perfectly clean.

6) The following may be checked as per approved drawing.
   a) Marking and cutting
   b) Correct sizes of dies
   c) Centering & fixing of sleeves
   d) Fixing of all the components i.e., Aluminium end pipes, hole plugs etc.
   e) Compression of sleeves at specified pressure.

7) All the sharp edges have been filled after compression.

8) There is no crack, bend or any damage to joint after compression.

FOR CONTRACTOR

Signature.....

Name.....

Designation.....

Date.....

FOR APTRANSCO

Signature.....

Name.....

Designation.....

Date.....

ANNEXURE CF IX
(5)

Dimensions for Mid Span Joints for 7/3.15 mm GSS Earthwire

1) Loc.No.

2) Span……Loc Nos……to…….

3) Approved Drg.No.

4) Details of Dimensions

Steel Portion						Aluminum Portion									
			As per Drg.			Actual					As per Drg.			Actual	
Before Comp.			After Comp.			After Comp.			Before Comp		After Comp.			After Comp.	
Length		Outer dia	Length	C-C	F-F	Length	C-C	F-F	Length	Outer dia	Length	C-C	F-F	Length	C-C	F-F

C-C : Corner to Corner Distance
F-F : Face to Face Distance

5) Bores in the sleeves are perfectly clean.

6) The following may be checked as per approved drawing.

   a) Marking and cutting
   b) Correct sizes of dies
   c) Centering & fixing of sleeves
   d) Fixing of all the components i.e., Aluminium end pipes, hole plugs etc.
   e) Compression of sleeves at specified pressure.

7) All the sharp edges have been filled after compression.

8) There is no crack, bend or any damage to joint after compression.

FOR CONTRACTOR

Signature.....

Name.....

Designation.....

Date.....

FOR APTRANSCO

Signature.....

Name.....

Designation.....

Date.....

NAME OF THE CONTRACTOR	DRUM SCHEDULE (CONDUCTOR) (SHEET 1-A)	ERECTION UNIT DATE SHEET NO.
NAME OF THE LINE		

SECTION From……to……

SECTIONAL LENGTH……

LOC.NO.
T.TYPE
SPAN _____ _____ _____ _____ _____ _____ _____

PHASE 0_____0_____0_____0_____0_____0_____0_____

0_____0_____0_____0_____0_____0_____0_____

0_____0_____0_____0_____0_____0_____0_____

0_____0_____0_____0_____0_____0_____0_____

0_____0_____0_____0_____0_____0_____0_____

0_____0_____0_____0_____0_____0_____0_____

0_____0_____0_____0_____0_____0_____0_____

SITE ENGINEER/APTRANSCO SITE INCHARGE/SUPERVISOR

NAME OF THE CONTRACTOR			DRUM SCHEDULE (CONDUCTOR) (SHEET 1-A)				ERECTION UNIT DATE SHEET NO.		
NAME OF THE LINE									
Sl.No.	Drum No.	Make	Length (Mtrs.)	Phase	Paying out between Locs	Quantity Consumed	Balance Bit Length	Bit to be used at	Remarks

NAME OF THE CONTRACTOR	DRUM SCHEDULE (CONDUCTOR) (SHEET 1-A)	ERECTION UNIT DATE SHEET NO.

NAME OF THE LINE

SECTION From.....to.....

SECTIONAL LENGTH.....

LOC.NO.
T.TYPE
SPAN _____ _____ _____ _____ _____ _____ _____

PHASE 0_____0_____0_____0_____0_____0_____0_____

0_____0_____0_____0_____0_____0_____0_____

Sl.No.	Drum No.	Make	Length (Mtrs.)	Phase	Paying out between Locs	Quantity Consumed	Balance Bit Length	Bit to be used at	Remarks

SITE ENGINEER/APTRANSCO SITE INCHARGE/SUPERVISOR